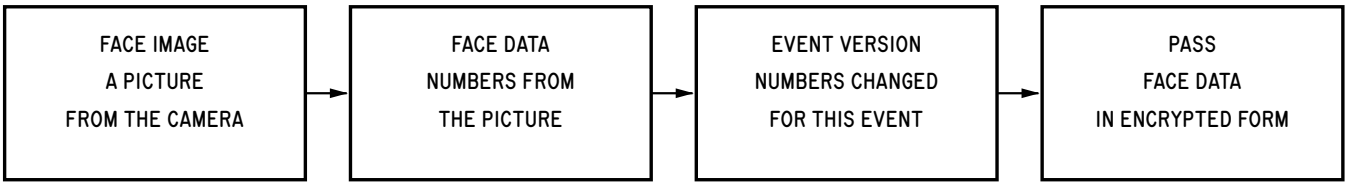
SECTION 1
FACE DATA AND MATCH ERRORS

FACE WORDS / MATCH ERRORS: PAGE 1 · PASS SECURITY / OFFLINE / TEST CLAIMS: PAGE 2

1.1 WHAT IS FACE DATA?

A face image is a picture. Face data is a set of numbers that a computer makes from that picture. A face template is another name for this set. Private Pass changes the numbers for each event before it puts them in a pass (figure 1).



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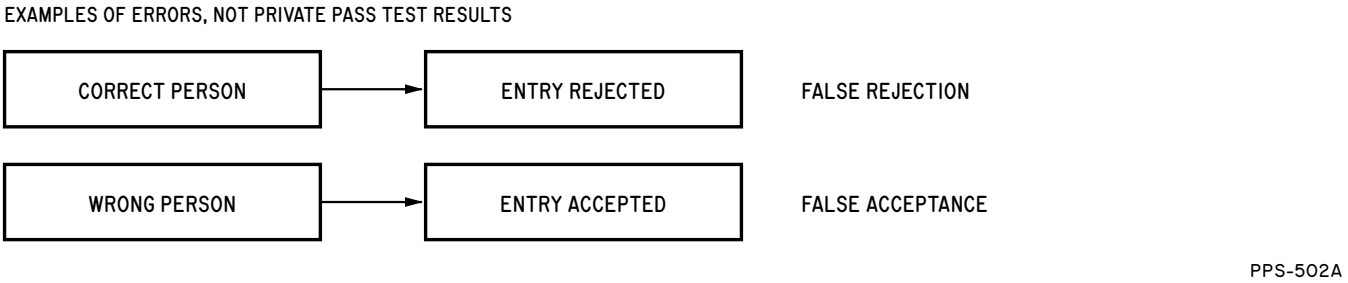
Figure 1. A Picture Becomes Data in a Pass

1.2 WHAT DOES 1:1 MEAN?

A 1:1 check compares one person with one pass. Example: Alex shows a pass. The gate compares Alex's face with the face data in that pass. It does not search a list of all visitors.

1.3 WHAT DOES THE MATCH LIMIT DO?

The match limit sets how close the two sets of numbers must be. With a small permitted difference, the gate can reject the correct person. With a large permitted difference, it can accept the wrong person. These are different errors (figure 2).



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Figure 2. Two Possible Errors in a Face Check

Table 1. Face Terms: Quick Lookup

Term in the talk	Meaning
Pass	A digital ticket with face data.
Gate	The device for the entry check.
Enrollment	Connects a person's face to a pass.
Template	Numbers made from a face image.
Embedding	Another name for a template.
Spoof attack	An attempt to enter with a fake face.

NOTE

Face data can show private information about a person. Numbers do not remove every privacy risk.

FIND AN EXPLANATION

1 Find the term in a heading.
2 Read the short explanation.
3 Use the example or drawing.
4 Use the table for other terms.

SECTION 2
PASS SECURITY AND OFFLINE ENTRY

FACE WORDS / MATCH ERRORS: PAGE 1 · PASS SECURITY / OFFLINE / TEST CLAIMS: PAGE 2

2.1 SIGNED AND ENCRYPTED: WHAT IS THE DIFFERENCE?

A digital signature lets the gate examine the source of a pass and find changes to it. Encryption changes face data into a form that only a prepared gate can read. A digital signature gives no encryption. A copied pass can still have a correct digital signature. The gate must also compare the face with the pass (figure 3).

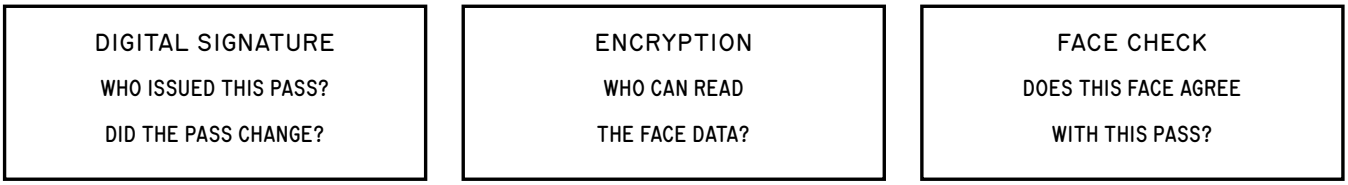


Figure 3. Each Check Answers a Different Question

2.2 HOW CAN THE GATE OPERATE OFFLINE?

Offline means no internet connection. The gate first downloads event data while it has an internet connection. It can then use the saved data for the entry decision. The server is the central computer for accounts and ticket records. The design does not keep face data here for use across different events.

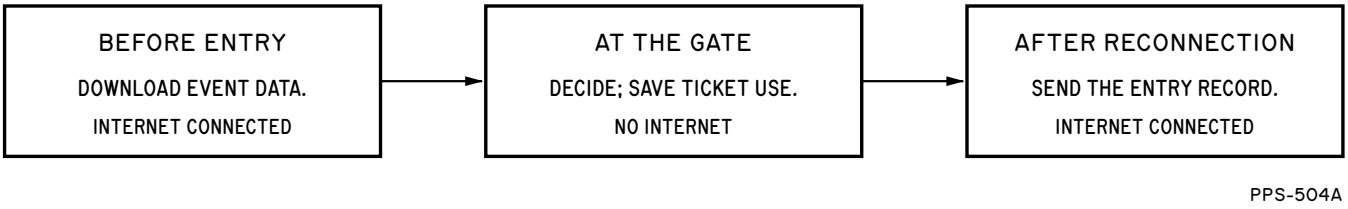


Figure 4. Offline Means Local Decisions, Then a Later Update

2.3 WHY CAN A CANCELED TICKET STILL PASS?

Example: A gate goes offline. The organizer then cancels a ticket. The gate cannot receive that change until it connects again. Offline entry uses the information already on the gate.

Table 2. How to Read a Test Claim

Phrase in the talk	What it establishes
Software test	A check of a specified part under test conditions.
Observed run	Someone saw the system operate. This alone gives no repeat test record.
Prototype	A test system. This word gives no performance result.

NOTE

A detected face can be a photo or a screen. Face detection alone does not stop a spoof attack.

EXAMINE A TEST CLAIM

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1 Find what the test measured.
2 Find the test conditions.
3 Look for a saved test record.
4 Look for more than one test.
```

Based on PrivatePass_EPQ Presentation_v4.pptx and its speaker notes. Examples explain concepts; they are not recorded test results.